Implementation and Validation of 1D Return Mapping Algorithms for Plasticity and Viscoplasticity

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Abstract

Constitutive material models are the cornerstone of finite element analysis in solid mechanics. Material behavior has direct consequences on how displacements, velocities and accelerations are calculated at the nodes of each elements. For viscoplastic behavior, mathematical definitions ties the evolution of the viscoplastic strain to the evolution of stresses and strain over time. Thus for each increment of strain and time, algorithm must find the equilibrium at wich Karush-Kuhn-Tucker (KKT) conditions are met. However, since viscoplastic material exhibit stress dissipation, these conditions needs to be relaxed to accomodate the accumulation of viscoplastic strain. In this mock paper, a return mapping algorithm for the Perzyna viscoplastic model will be implemented to solve for the equilibrium of the KKT conditions considering the overstress concepts. The basic definitions of return mapping for regular plasticity will be showed and will serve as a basis for further implementation of the Perzyna model to show the impact of strain rates and material properties on constitutive behavior.

Keywords: Return mapping, Plasticity, Viscoplasticity, Perzyna model, Backward Euler

1. Introduction

Consitutive material models are required to model the material response to loads in finite element analysis. For solid mechancis, variable of interest are the displacement, velocity and acceleration of each and every node to compute derived variables like stresses and strains. For linear behavior, where the stress response follows a linear relationship with strains (by definition displacements) this problem is rather easy to solve but even then, advanced algorithm are used to determine the displacements, stresses and strains efficiently. The same goes for other material behavior like plasticity and viscoplasticity.

Return mapping is a well known algorithm and have been used for well over forty years. Early work by [1] on radial return have paved the way to the work of [2]. Since then, great textbook have also covered the topic and applied it to several material models (consitutive model)[3] [4].

The goal of this mock paper is to condense gathered knowledge about computational plasticity, especially about return mapping applied to 1D viscoplastic constitutive Perzyna model. First, a survey of the theoretical background of return mapping will be presented. Then, the implementation of the algorithms will be shown and finally key results and observation to common material will be studied.

2. Theoritical background

The governing equations in solid mechanics are defined by:

$$\nabla \cdot \boldsymbol{\sigma} + \boldsymbol{b} = \rho \ddot{\boldsymbol{u}} \tag{1}$$

$$\sigma_{ij} = \mathbb{C}_{ijkl}\varepsilon_{ij} \tag{2}$$

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{3}$$

known as the equilibrium equation (1), the constitutive law (2) and the strain-displacement relation (3).

In a purely linear, 1D isotropic case (2) can be simply seen as the product of the Young's modulus (E) and the strain applied to a rod (ε). In multidimensional nonlinear computational mechanics, this can also be seen as the *trial stress* or *elastic predictor*. As introduced in the last sentence, return mapping starts with an initial guess to see if the new increment of strain generate enough stress so that plasticity or viscoplasticity is indused in the material. To consider that inelastic stress is induced, the *elastic predictor* must give a value that is beyond elastic stress, thus a value larger than σ^y. The stress from a loading is given by a yield function (e.g. Von Mises), and the concept of if the stress state is elastic or inelastic can be verified with the yield criterion ($f(\boldsymbol{\sigma})$). The yield criterion takes at its most simple form, the stress state at the next step ($n+1$) and the parameter σ^y and is defined as:

$$f(\boldsymbol{\sigma}_{n+1}) = \boldsymbol{\sigma}_{n+1} - \sigma^y_n \tag{4}$$

Then, three conditions has to be met at all times, the Karush-Kuhn-Tucker (KKT) conditions:

$$\gamma \geq 0 \tag{5}$$

$$f(\boldsymbol{\sigma}) \leq 0 \tag{6}$$

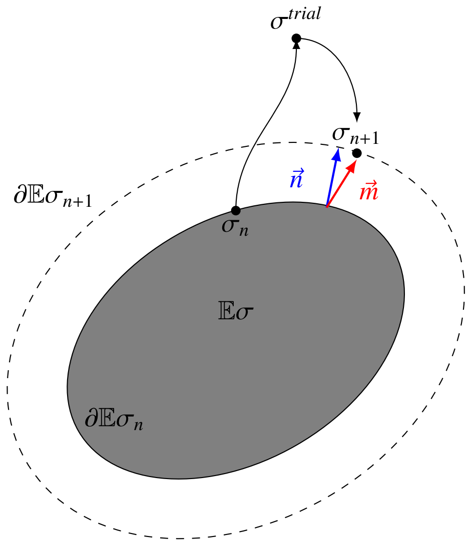

Figure 1: Return mapping in 1D with strain hardening

$$\gamma f(\boldsymbol{\sigma}) = 0 \tag{7}$$

Here, the symbol γ is called the plastic multiplier and is part of the return mapping algorithm. Essentially, the plastic multiplier scale the vector giving the direction of plastic flow (either $\vec{n}$ or $\vec{m}$) to give the inelastic strain increment (See Figure 1). Here $\vec{n}$ is the vector normal to the yield function and $\vec{m}$ the vector normal to the plastic potential. If $\vec{n} = \vec{m}$,meaning that the yield function and plastic potential are the same, this is called associative plasticity. The opposite case is called anassociative, when $\vec{n} \neq \vec{m}$. Typically, viscoplasticity is associative [4, 3]. This normality condition thus defines the inelastic strain increment:

$$\dot{\boldsymbol{\varepsilon}}^{inelastic} = \gamma \boldsymbol{m} \tag{8}$$

Equations 5, 6, 7 constrain the problem so that for any stress state either $f(\boldsymbol{\sigma}) < 0$ meaning that the stress state is purely elastic or either $\gamma > 0$ meaning inelastic strain is increasing. One other important aspect is that $f(\boldsymbol{\sigma})$ must at most be equal to zero. As seen in Figure 1, an inital stress state at step (n) is defined by a domain of admissible stress ($\mathbb{E}\sigma$) and a boundary to this domain ($\partial\mathbb{E}\sigma_n$). The way the next stress state is found is, as stated before, when all KKT conditions are met.

2.1. Rate-Independent Plasticity

Rate-independence means that what ever the rate the material is deformed, properties do not change because of it. This however, does not mean that material cannot undergo strain hardening. In this case, it is quite straightforward to obtain the increment of plastic strain. For the sake of simplicity and stay in line with the mock paper objectives, only 1D case will be defined for perfect plasticity ($H = 0$) and isotropic hardening ($H > 0$).

Knowing that the elastic predictor is:

$$\sigma^{trial} = \sigma_n + E\Delta\varepsilon \tag{9}$$

we combine it to the extension of equation 4 to obtain the definition of the next step stress:

$$\sigma_{n+1} = \sigma^{trial} - E\Delta\varepsilon^p \tag{10}$$

from the equation 8, and assuming 1D plasticity we then obtain:

$$\sigma_{n+1} = \sigma^{trial} - E\Delta\gamma sign(\sigma^{trial}) \tag{11}$$

and finally, we rearrange the terms to obtain the increment in plastic multiplier:

$$\Delta\gamma = \frac{f^{trial}}{E} \tag{12}$$

Back to the normality equation 8, the increment in plastic strain is obtained. Also, similar approach is taken to define the increment of plastic multiplier for plasticity with hardening. Then the increment increment is:

$$\Delta\gamma = \frac{f^{trial}}{E + H} \tag{13}$$

2.2. Viscoplasticity

In some cases and conditions, metals exhibit viscoplastic behavior. Viscoplasticity can be defined by the dependence of material state evolution to the strain rate. This behavior can be represented by a spring-frictional-dashpot element see Figure 2. The dashpot add a relationship with the strain rate and is defined by a viscosity property.

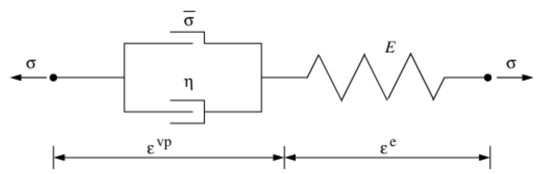

Figure 2: Spring-frictional-dashpot element representation (from [4])

The definition of the viscoplastic strain rate ($\dot{\varepsilon}^{vp}$) in the Perzyna model is:

$$\dot{\boldsymbol{\varepsilon}}^{vp} = \frac{1}{\eta} \langle \varphi(f) \rangle \boldsymbol{m} \tag{14}$$

where:

$$\varphi(f) = \left(\frac{f}{\bar{\sigma}}\right)^n \tag{15}$$

The parameters n and η change the behavior of the material:

- n: rate sensitivity exponent
- η: viscosity parameter

Similar development as in section 2.1 and in the relationship between $\dot{\varepsilon}^{vp}$ and γ is made. The exception is since it is a time dependent solution, the variables must be integrated. Typically, a backward Euler scheme is used for unconditional stability because it is fully implicit (See [4, 3]). The definition of the increment of the plastic multiplier, for the 1D case is then (in incremental form):

$$\Delta\gamma = \frac{\Delta t}{\eta} \left\langle \frac{f}{\bar{\sigma}} \right\rangle^n \tag{16}$$

For the Perzyna model, the KKT conditions are relaxed, meaning that $f > 0$ is allowed. The reason is that the viscous stress (or over stress) is considered as a complementary state and not included implicitely in the definition in the stress state of the material point since it fades over time. The only limiting case for the Perzyna model is when $\eta \rightarrow 0$ as this solution recovers the rate-independant solution.

The Perzyna model is relatively simple to implement and requires few parameters (minimize the number of tests to calibrate the material model). However, other models and frameworks exist and take into account other mechanics and/or parameters. Here, parameters are identified through the mean of optimization, identification algorithm or any other numerical method (e.g. Levenberg-Marquardt algorithm, genetic algorithm, machine learning, etc.). To name a few:

- Duvaut-Lions: add a back-bone stress concept ($\bar{\sigma}$). Viscoplastic strain is then computed as:

$$\dot{\boldsymbol{\varepsilon}}^{vp} = \frac{1}{\tau} \left(\boldsymbol{D}^{e}\right)^{-} 1(\boldsymbol{\sigma} - \bar{\boldsymbol{\sigma}}) \tag{17}$$

  where τ is the relaxation time and $\boldsymbol{D}^{e}$ the elastic stiffness matrix linked to material properties. Back-bone stress is the stress state in the rate-independent case. Meaning that on every iteration, the rate-idependent case has to be solved along the viscoplastic state.

- Consistency model: The stress state must lie on the yield surface. Thus the redefinition of equation 6 gives:

$$f(\boldsymbol{\sigma}, \dot{\boldsymbol{\varepsilon}}^{vp}) \leq 0 \tag{18}$$

- Johnson-Cook [5]: An empirical model, meaning it is based on paramete calibration rather than material thermodynamics, with the form:

$$\sigma_y = \underbrace{(A + B\varepsilon_p^n)}_{\text{strain hardening}} \cdot \underbrace{(1 + C \ln \dot{\varepsilon}^*)}_{\text{strain rate sensitivity}} \cdot \underbrace{(1 - T^{*m})}_{\text{thermal softening}} \tag{19}$$

  The advantages of the Johnson-cook model is that it is the yield stress that is defined through the model, therefore the stress can be evaluated with any other yield function. An other advantage is the modularity. If for example the material do not exhibit strain hardening or thermal softening, the equation becomes:

$$\sigma_y = A + C \ln \dot{\varepsilon}^* \tag{20}$$

- Chaboche [6]: Framework used in cyclic viscoplasticity. It includes:
  - Isotropic hardening
  - Multiple backstress tensors with Armstrong-Frederick kinematic hardening rules
  - Viscous flow rule (Typically Norton-type or Perzyna-type)

3. Implementation

In this section, the implementation details of the return mapping algorithm applied to first, the elastoplastic model with linear hardening, and then for the Perzyna-type viscoplastic modelin 1D. The extansion to higher dimension is pretty straightforward.

3.1. Algorithm Structure and Code Organization

The algorithm, for either models, have the same basic structure:

- Elastic predictor
- Yield check
- Inelastic corrector

In the case of the linear hardening elastoplastic model, the application is pretty straightforward in regard to the previous list (see Algorithm 1)

Algorithm 1: LinearHardeningReturnMapping Solve the stress state for next stress increment using return mapping for the linear hardening elasto-plastic model

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Input: Current step material data $(\sigma_n, \varepsilon_n, \varepsilon_n^p, \sigma^{y0}, \alpha_n, H)$
Output: Updated material data $(\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^p, \alpha_{n+1})$
$\sigma_n^y \leftarrow \sigma^{y0} + H\alpha_n$;
$f(\boldsymbol{\sigma}^{trial}) \leftarrow \sigma_n + E(\Delta\varepsilon - \varepsilon_n^p) - \sigma_n^y$;
if $f(\boldsymbol{\sigma}^{trial}) < 0$ **then**
  $\sigma_{n+1} \leftarrow \sigma^{trial}$;
  $\varepsilon_{n+1} \leftarrow \varepsilon_n + \Delta\varepsilon$;
  $\varepsilon_{n+1}^p = \varepsilon_n^p$;
else
  $\Delta\gamma \leftarrow \dfrac{f(\sigma^{trial})}{E + H}$;
  $\Delta\varepsilon_p \leftarrow \Delta\gamma \cdot sign(\sigma^{trial})$;
  $\varepsilon_{n+1}^p \leftarrow \varepsilon_n^p + \Delta\varepsilon^p$;
  $\alpha_n \leftarrow \alpha_n + \Delta\gamma$;
  $\sigma_{n+1} \leftarrow \dfrac{1 - \Delta\gamma E}{|\sigma^{trial}|}\sigma^{trial}$;
end
return $\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^p, \alpha_{n+1}$;

In the case of the Perzyna-type viscoplastic model, the implementation is similar. It only incorporates a nonlinear solve if the strain-rate sentitivity parameter $n > 1$ (See algorithm 2)

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Algorithm 2: PERZYNARETURNMAPPING Solve the stress state for next stress increment using return mapping for the Perzyna-type viscoplastic model

Input: Current step material data ($\sigma_n, \varepsilon_n, \varepsilon_n^{vp}, \eta, n$)
Output: Updated material data ($\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^{vp}$)
$f(\sigma^{trial}) \leftarrow \sigma_n + E(\Delta\varepsilon - \varepsilon_n^{vp})$;
if $f(\sigma^{trial}) < 0$ **then**
  $\sigma_{n+1} \leftarrow \sigma^{trial}$;
  $\varepsilon_{n+1} \leftarrow \varepsilon_n + \Delta\varepsilon$;
  $\varepsilon_{n+1}^{vp} = \varepsilon_n^{vp}$;
else
  if $n = 1$ **then**
    $\Delta\gamma \leftarrow \dfrac{\Delta t f(\sigma^{trial})}{\eta \Delta t E}$
  else
    $\Delta\gamma \leftarrow NonLinear\Delta\gamma(f(\sigma^{trial}), \Delta t, E, \sigma^y, \eta, n)$;
  end
  $\varepsilon_{n+1}^{vp} \leftarrow \varepsilon_n^{vp} + \Delta\gamma \cdot sign(\sigma^{trial})$;
  $\sigma_{n+1} = \sigma^{trial} - E\Delta\gamma \cdot sign(\sigma^{trial})$
end
return $\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^{p}, \alpha_{n+1}$;

Algorithm 3: $NonLinear\Delta\gamma$ Solve the plastic multiplier increment for the Perzyna-type viscoplastic model for the $n > 1$ non-linear case

Input: $f(\sigma^{trial}), \Delta t, E, \sigma^y, \eta, n$
Output: $\Delta\gamma$
$itermax \leftarrow 30$;
$tol \leftarrow 10^{-10}$;
return $\Delta\gamma$;

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